Online Assignment 1

Duration: 90 minutes

Question 1 (60 points): TCP Socket Programming – Number Prediction

Write a **Client** and a **Server** program using Java TCP Socket programming to implement the following:

1. **Server Logic:** Whenever a Client send a connection request to Server, it accepts this client and generates a random integer R between 0 and 99 for prediction.
2. **Client Logic:** Upon connecting, the Client allows the user to input an integer and sends it to the Server.
3. **Comparison:** The Server compares the received number with R and returns one of the following messages:
   - If the number equals R: **"Congratulations! You predicted correctly."**
   - If the number is smaller than R: **"Your number is smaller. Please predict again."**
   - If the number is bigger than R: **"Your number is bigger. Please predict again."**
4. **Termination:** If the user at Client side predicts correctly, the Client program should be stopped and the connection at Client side and Server side should be closed.

Note: To generate a random integer in the range [0, 99] in Java, use:

```
int R = (int) (Math.random() * 100);
```

a. Implement the Client and Server programs to support **only one** client. (40 points)

b. Rewrite the Server program to support **multiple clients simultaneously by using thread pool**. The Server must be restricted to a maximum of 20 concurrent clients at any given time. Each client should have its own unique number R to predict. (20 points)

How to submit: you need to submit two projects: 1 project for a) and 1 project for b)

Question 2 (40 points): (UDP Socket programming) Sensor monitoring system

Write Java UDP client and server program for a smart home sensor monitoring system in which the server supports only one client. The client allows the user to enter **<u>an arbitrary number</u>** of sensor readings in any order and sends them to the server through UDP. The supported readings are room temperature (float), humidity (integer), and door status (string). Since UDP does not guarantee packet ordering, the server may receive the readings out of order. Your program must ensure that, after transmission is completed, the server can correctly interpret every received reading and display the full list in the same order as the user entered the data. Assume that no packet is lost and no packet is duplicated. When the user finishes entering data, the client must

send a special message to inform the server that no more readings will be sent. Design the packet format yourself.

Hint: At the Client side, the Client should display the following menu for a user:

1. Room temperature
2. Humidity
3. Door status
4. Exit

The user can choose an arbitrary number, enter the value and send this value to Server immediately. **Each packet only contains one value of one information**. If user choose 4, a special packet END is sent to server and the client program will be stopped.

Examples:

Example 1
At the client side, the user enters the following readings in this order and send 4 separate packets:

- room temperature: 26.5
- humidity: 60
- door status: OPEN
- END message

Even if the packets arrive at the server in a different order, the server must finally display:

- room temperature: 26.5
- humidity: 60
- door status: OPEN

Example 2
At the client side, the user enters the following readings in this order and send 5 separate packets:

- door status: CLOSED
- room temperature: 29.0
- humidity: 72
- door status: OPEN
- END message

Even if the server receives these readings out of order, it must finally display:

- door status: CLOSED
- room temperature: 29.0
- humidity: 72
- door status: OPEN

a) Design an appropriate packet format for communication between client and server. And you need to briefly explain how your packet design allows the server to correctly interpret each reading and restore the original input order.
b) Write Client program and Server program.

How to submit: you need to submit 1 file (text file) and 1 project: file for a) and 1 project for b).